

Why crop rankings do not determine land allocation

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Stage II final scientific draft

Abstract

Crop rankings are ordinal, whereas acreage allocation is a cardinal decision made under joint uncertainty and operational constraints. We formalize this gap with a set-valued, multi-crop programme and distinguish possible, universal and selected ranking reversal. Exact results show that rankings alone do not identify acreage and that operational feasibility can force universal reversal even when downside risk is inactive. In pre-specified simulations, E2 operational interventions convert a corn-only allocation into soybean-majority allocations in every treated cell; all 24/24 family-wise intervals meet the precision criterion. E6 shows that the interaction between information and action-set flexibility can be positive (3.258, 95% interval 2.788 to 3.728), exactly null or substitutive. E1, E3, E4 and E5 do not meet their experiment-level precision criteria and remain inconclusive. An official US panel extending from 2016 to 2024 contains 248 complete state-years in 31 states. Concurrent operating-margin inversion intensity is 0.411 (state-cluster 95% interval 0.346 to 0.475), but all four strictly lagged acreage-share intervals include zero and national summaries depend on aggregation. Ranking reversal is therefore demonstrable under explicit mechanisms, while public acreage disagreement alone does not identify private objectives, constraints, risk preferences, dependence, causality or welfare.

1 Introduction

Crop prediction systems increasingly turn weather, soil, management and remote-sensing data into yield forecasts, suitability scores and crop rankings [Liakos et al., 2018, van Klompenburg et al., 2020]. These outputs answer an ordinal question—which crop scores higher—but not the operational question of how much acreage to assign to each crop. A ranking remains unchanged under every strictly increasing transformation of its scores; an acreage allocation generally does not. Moving from prediction to prescription therefore requires cardinal margins, a joint uncertainty model, operational constraints and a rule for selecting among multiple optima.

Crop-planning studies represent acreage as a constrained portfolio decision with resource limits, rotations, stochastic returns, robustness or downside risk [Filippi et al., 2017, Boyabath et al., 2019, Benini et al., 2023, Randall et al., 2022, 2024]. Prescriptive analytics and decision-focused learning likewise separate predictive accuracy from downstream decision quality [Bertsimas and Kallus, 2020, Elmachtoub and Grigas, 2022, Wilder et al., 2019, Mandi et al., 2023]. Yet a score-acreage mismatch alone does not reveal whether a decision is constrained, risk-sensitive, inefficient or even unique. The same visual pattern can arise from different mathematical mechanisms.

Here we reconstruct the ranking-to-allocation argument as a closed evidence chain (Fig. 1). A set-valued stochastic programme maximizes expected margin over a polyhedral operational set under loss-CVaR, using the atom-safe representation of Rockafellar and Uryasev [2000, 2002]. Reversal can occur at some optimum, every optimum or only the selected optimum. Exact constructions separate cardinal-margin, operational, downside-risk and multiplicity mechanisms, while a complete Karush–Kuhn–Tucker identity records local pressure without converting marginal conditions into acreage ranks.

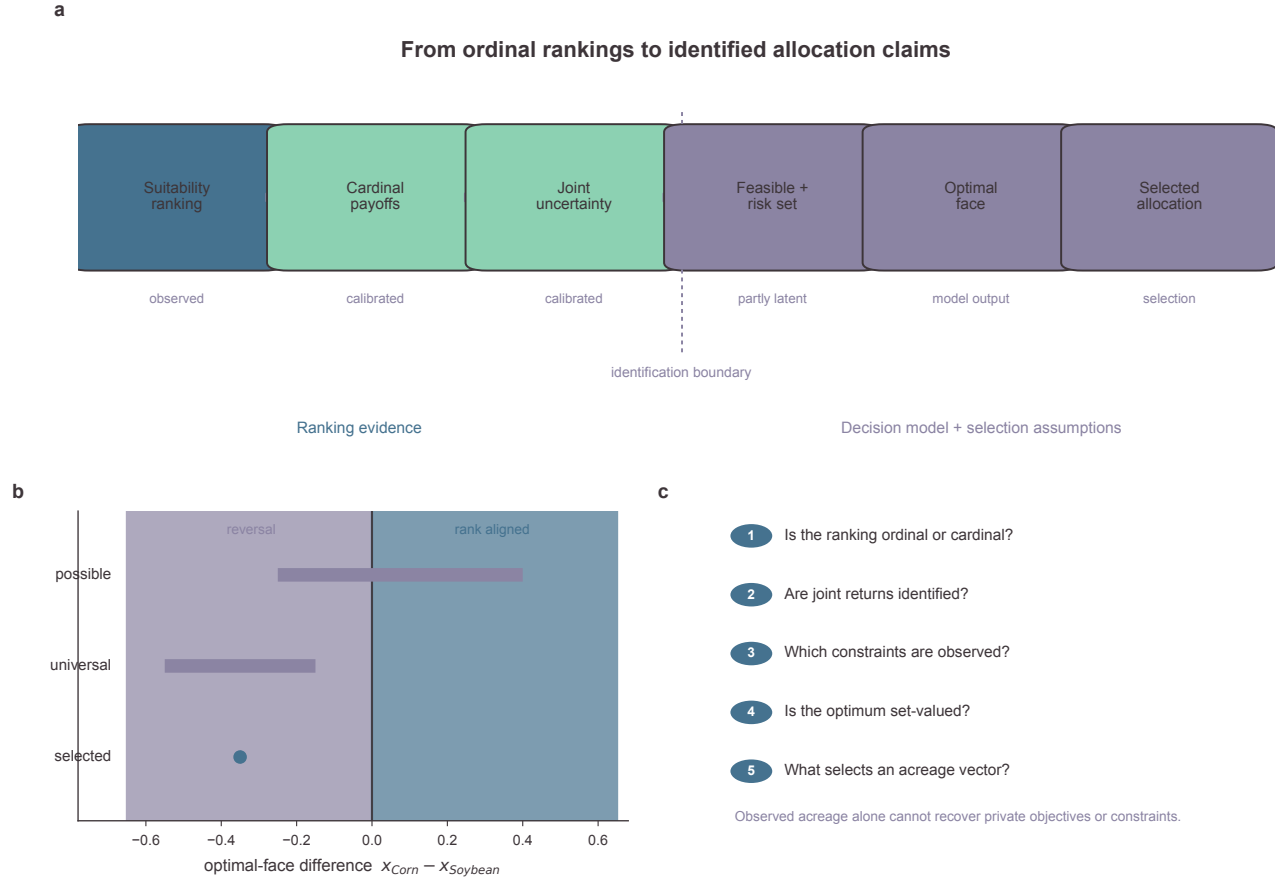


Figure 1: From crop ranking to an allocation claim. Cardinal payoffs, joint uncertainty, an operational feasible set and a selection rule intervene between an ordinal ranking and an acreage vector. Possible and universal reversal are properties of the full optimal face; selected reversal depends on a declared selection rule.

Six pre-specified experiments test this mechanism framework. Only E2 and E6 meet every experiment-level precision criterion. E2 assigns operational interventions and produces universal reversal throughout the treated cells. E6 crosses signal information with action-set flexibility and yields positive, exact-null and substitutive interactions across three constructive environments. E1, E3, E4 and E5 reach their replication ceilings without meeting every registered interval criterion; their complete results remain visible in the Supplementary Information but do not support positive claims.

The empirical analysis asks a narrower question using official observations. The reconstructed panel covers 2016–2024, 248 complete state-years and 31 states. Concurrent score–acreage inversion is definition- and state-dependent, whereas all four strictly lagged acreage-share intervals include zero and national aggregation changes the pattern. These observations document descriptive disagreement, not farm objectives or optimal acreage. The contribution is consequently both positive and bounded: explicit mechanisms can generate ranking reversal, but public rank–acreage disagreement cannot select among those mechanisms without decision-level information.

2 Ranking does not identify allocation

A crop ranking and an acreage allocation are different mathematical objects. Let a suitability vector s induce a weak order and let $x \in \mathbb{R}_+^n$ denote acreage. Strictly increasing transformations preserve the

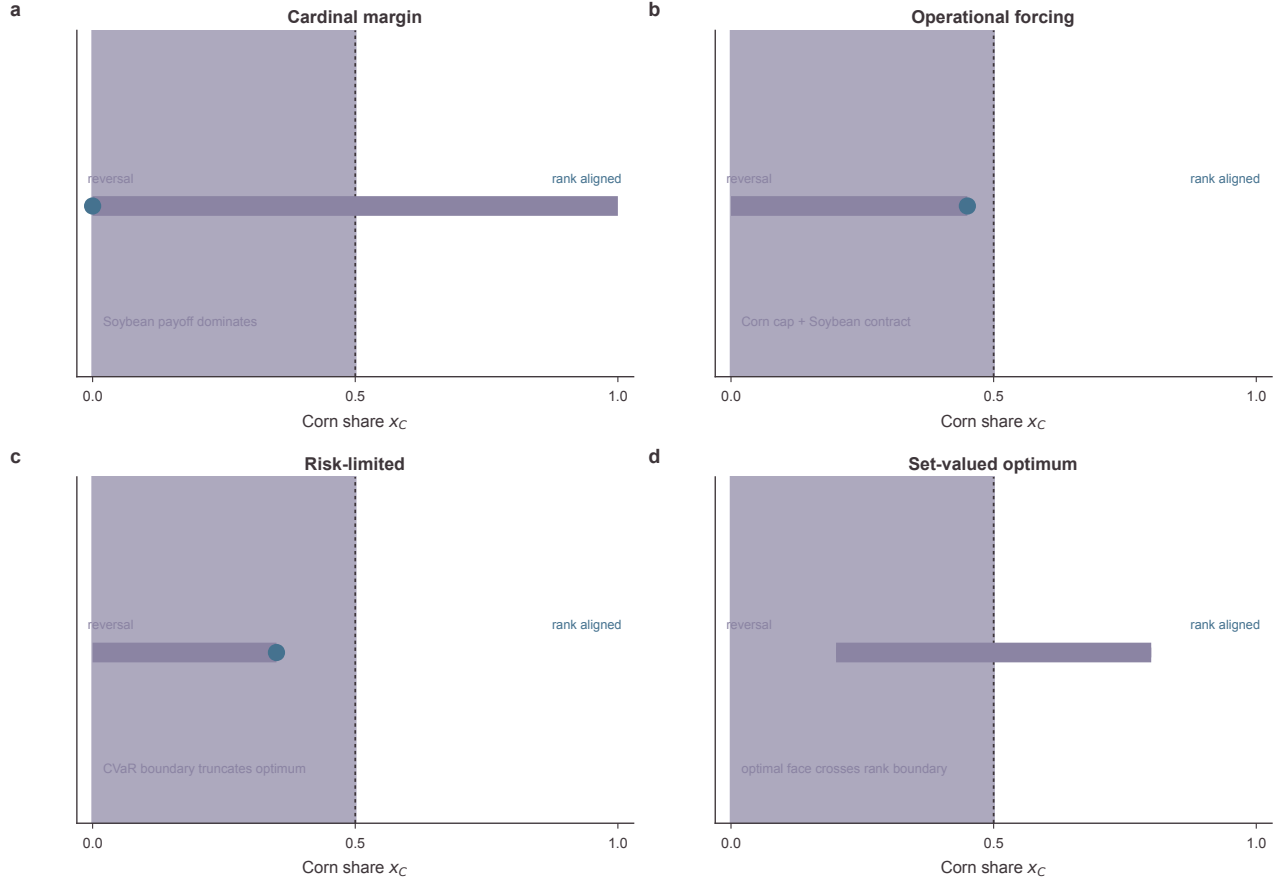


Figure 2: A four-mechanism atlas on shared coordinates. Cardinal-margin, operational, downside-risk and set-valued cases use the same corn-share axis and reversal boundary. The cases are exact analytic constructions, not empirical estimates.

order in s but can change any proportional score-to-acreage rule. The ranking also supplies neither an objective nor a feasible set. Consequently, even a strict order cannot identify a cardinal acreage vector.

The distinction becomes sharper when the optimum is set-valued. Let

$$X = \{x : \ell \leq x \leq u, \mathbf{1}^\top x \leq A, c^\top x \leq B, Gx \leq g\} \quad (1)$$

and let $r_\alpha(x) = \text{CVaR}_\alpha[-\tilde{\pi}^\top x]$ be loss-CVaR. The optimal face is

$$S_\kappa = \arg \max \{\mu^\top x : x \in X, r_\alpha(x) \leq \kappa\}. \quad (2)$$

For a ranked pair $s_i > s_j$, reversal is *possible* if $x_j > x_i$ at some $x \in S_\kappa$, *universal* if it holds at every $x \in S_\kappa$, and *selected* if it holds at the allocation returned by a declared deterministic rule. These definitions prevent one solver vertex from being mistaken for the entire optimal face.

Four exact two-crop constructions share the same coordinate system (Fig. 2). Cardinal margins can point against the external crop ranking. A bound or contract can place the entire feasible set in the reversal half-space. A binding downside-risk constraint can move the optimum without implying monotone crop-wise comparative statics. A flat optimal face can cross the rank boundary, making reversal possible but not universal. Identical-looking allocations can thus encode different claims.

Operational feasibility supplies a transparent universal certificate. If $s_i > s_j$ and

$$\sup_{x \in X} x_i + \tau_x < \inf_{x \in X} x_j, \quad (3)$$

then every feasible allocation reverses the pair; $u_i + \tau_x < \ell_j$ is a simple sufficient condition. This result needs no downside-risk channel. Conversely, if an unconstrained expected-margin optimum is risk-feasible, the risk constraint leaves the optimal value unchanged, but one slack selected solution does not establish equality of optimal faces when optima are multiple.

The complete first-order identity blocks a second shortcut. At an optimum, non-negative multipliers and a subgradient $d \in \partial r_\alpha(x)$ satisfy

$$0 = -\mu + \lambda d + \gamma \mathbf{1} + \beta c + G^\top \eta - a + b. \quad (4)$$

For crops i and j ,

$$\mu_i - \mu_j = \lambda(d_i - d_j) + \beta(c_i - c_j) + (G^\top \eta)_i - (G^\top \eta)_j - (a_i - a_j) + (b_i - b_j). \quad (5)$$

The common land multiplier cancels, whereas budget, shared-constraint and bound terms remain. Equation 5 describes local marginal balance; it does not rank acreage levels.

The closed M0–M4 path makes these distinctions auditable (Fig. 3). Successive stages add cardinal margins, operations, downside risk and dependence to a common benchmark. Allocation and indexed outcome panels retain the actual registered outputs; an all-subset Shapley decomposition reports attribution intervals and closes with maximum efficiency residual 1.42×10^{-14} . Signed E2 pressure terms show how margin pressure is balanced locally by budget, shared or boundary terms, with maximum stationarity residual 1.82×10^{-11} . These are accounting decompositions over a controlled model, not causal effects in observed acreage.

3 Operational constraints identify reversal

E2 assigns budget, rotation, contract and dominant-crop-bound interventions while holding the ranking and scenario law fixed (Fig. 4). The base allocation assigns corn share 1.000. Each treated cell instead assigns a soybean majority: contract or rotation interventions produce corn and soybean shares 0.450 and 0.550, while the tighter budget produces 0.318 and 0.682. Possible, universal and selected reversal coincide in every treated cell, including the bound anchor.

All 24 of 24 registered family-wise intervals meet their precision targets at 16 replications per cell. The optimal-face classifications rule out selection dependence, and the local KKT audit shows that the loss-CVaR term is inactive in these cells. Because the intervention is assigned within a closed simulation, E2 identifies an operational displacement mechanism in its declared domain. It does not estimate how often that mechanism operates in agriculture.

The evidence rule is experiment-level rather than contrast-level. E1, E3, E4 and E5 each reach their registered ceiling without meeting every primary precision criterion. Passing rows inside those experiments do not rescue the experiment. E1 is reported in Supplementary Fig. S1, E3 in Supplementary Fig. S2, and E4/E5 in Supplementary Fig. S3. All 206 registered infeasible rows remain in the evidence ledger; no feasible-only subset enters a positive claim.

4 Information value depends on available actions

E6 crosses signal information with action-set flexibility in three constructive environments. Define the interaction as the difference-in-differences between informative and uninformative signals under flexible and restricted action sets. Ignoring a signal remains feasible, so information value is non-negative in each action set; exact ignore-signal anchors and signal-garbling checks verify this implementation invariant.

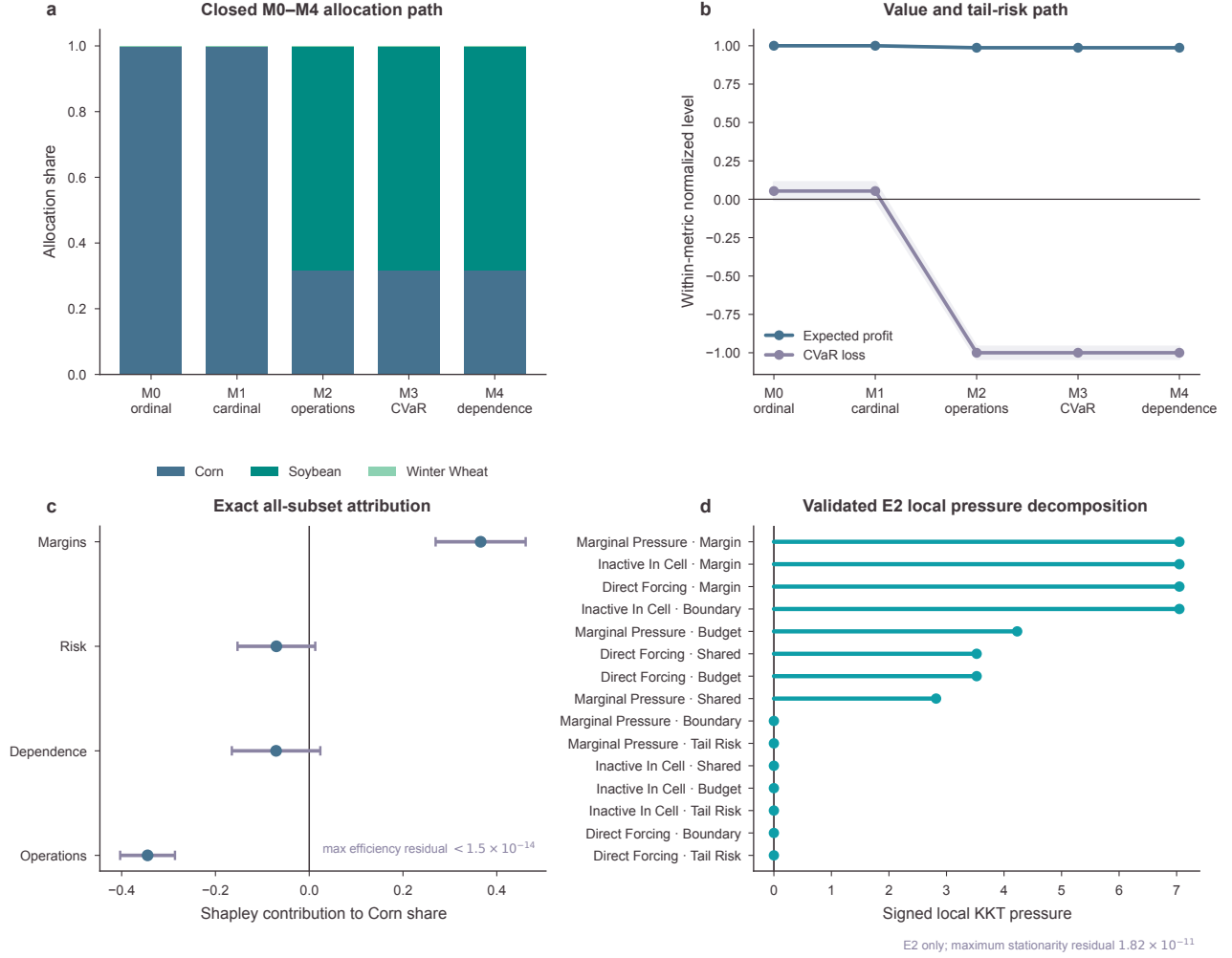


Figure 3: Nested allocation and mechanism decomposition. The M0-M4 path reports crop allocation, indexed outcomes, all-subset Shapley contributions and signed E2 KKT pressures. Intervals summarize registered closed-domain repetitions; pressure terms are local accounting identities rather than causal acreage effects.

The interaction spans all three theoretically admissible cases (Fig. 5). When flexibility unlocks state-contingent specialization, the estimate is 3.258 with 95% family-wise interval 2.788 to 3.728. A robust fallback makes information and flexibility substitutes, giving -3.114 with interval -3.476 to -2.751. A dominated added option yields the exact-null interaction 0.000. All three intervals meet the precision criterion at 16 replications.

This sign pattern shows why information and flexibility are not universally complementary. Information is actionable when it changes a feasible contingent decision, but the incremental value of the signal depends on what the enlarged action set adds. E5 cross-law dependence diagnostics are therefore kept separate in Supplementary Fig. S3: its experiment-level precision criterion was not met, so it cannot support a general claim about copula misspecification, regret or downside risk.

5 Empirical patterns are definition-dependent

The official-data reconstruction extends the common state panel to 2016–2024 (Fig. 6). It contains 744 complete state-crop-year rows, 248 complete state-years and 31 states. State acreage and yield come

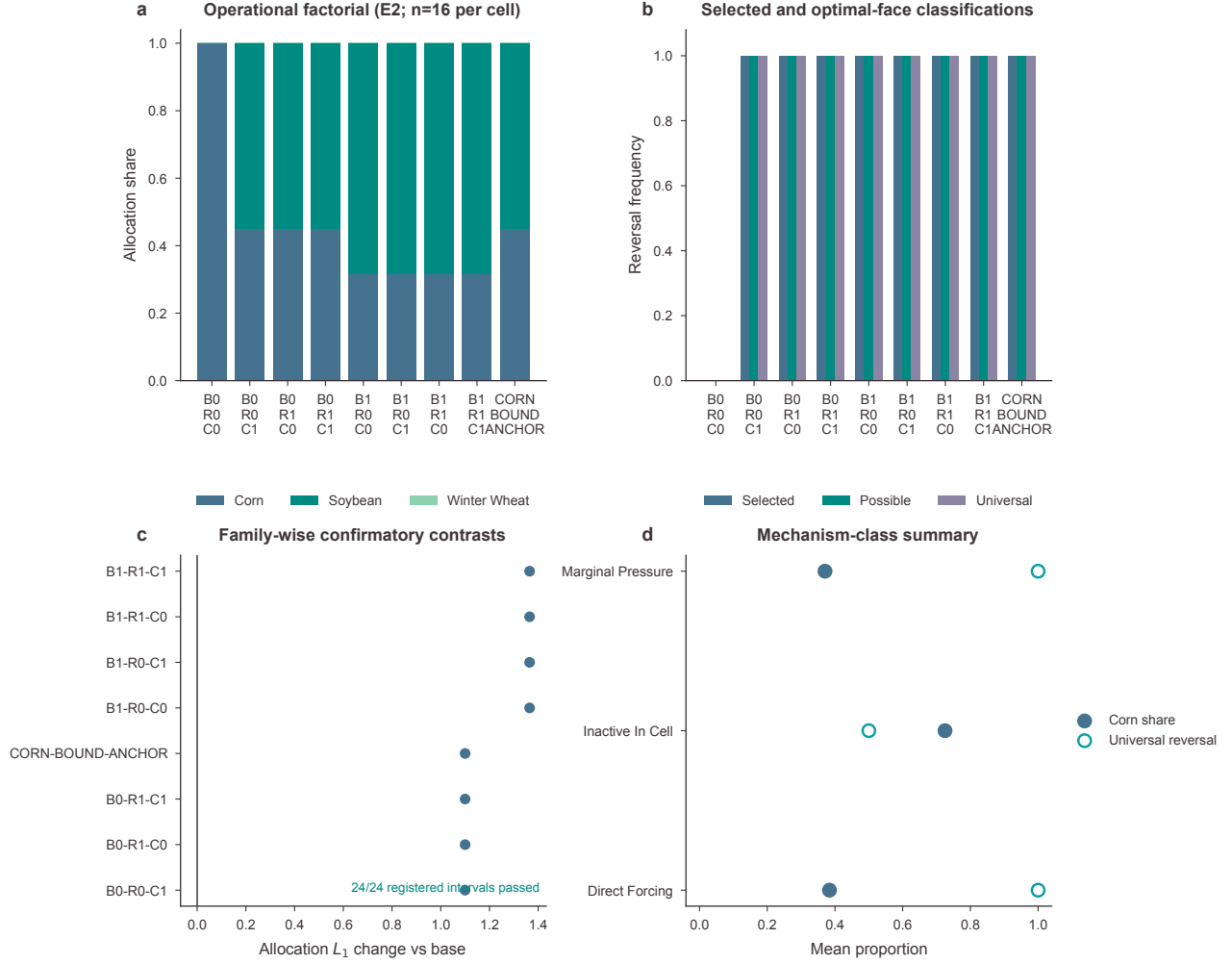


Figure 4: Operational interventions identify reversal in E2. The E2-only display reports the factorial allocation, selected and optimal-face classifications, family-wise contrasts and mechanism-class summaries. Each cell has $n = 16$ and all 24/24 intervals meet the pre-specified precision criterion.

from non-overlapping USDA National Agricultural Statistics Service annual-report blocks. National prices and costs come from USDA Economic Research Service series, and the registered CPI-U vintage converts monetary variables to real dollars. Missing or suppressed acreage and yield are not imputed.

Concurrent score and acreage orders differ materially. Operating-margin inversion intensity is 0.411 with a state-cluster 95% interval of 0.346 to 0.475, and its score leader differs from the acreage leader in 62.5% of complete state-years. Relative yield, standardized revenue and total-cost margin give different patterns, showing that an inversion statement inherits the accounting definition used to construct the score. The ranked state display further shows that the operating-margin pattern is not spatially uniform.

The strictly lagged analysis does not provide positive evidence that a previously leading score predicts a larger subsequent acreage-share change. For operating margin, the prior-leader coefficient is 0.0021 with interval -0.0041 to 0.0084. The other three primary intervals also include zero. Across the 217 transitions per definition, the acreage leader changes in 24 operating-margin transitions while score-leader movement is definition-specific. This is compatible with persistence, but the aggregate data do not measure rotations, adjustment costs, contracts or farmer expectations.

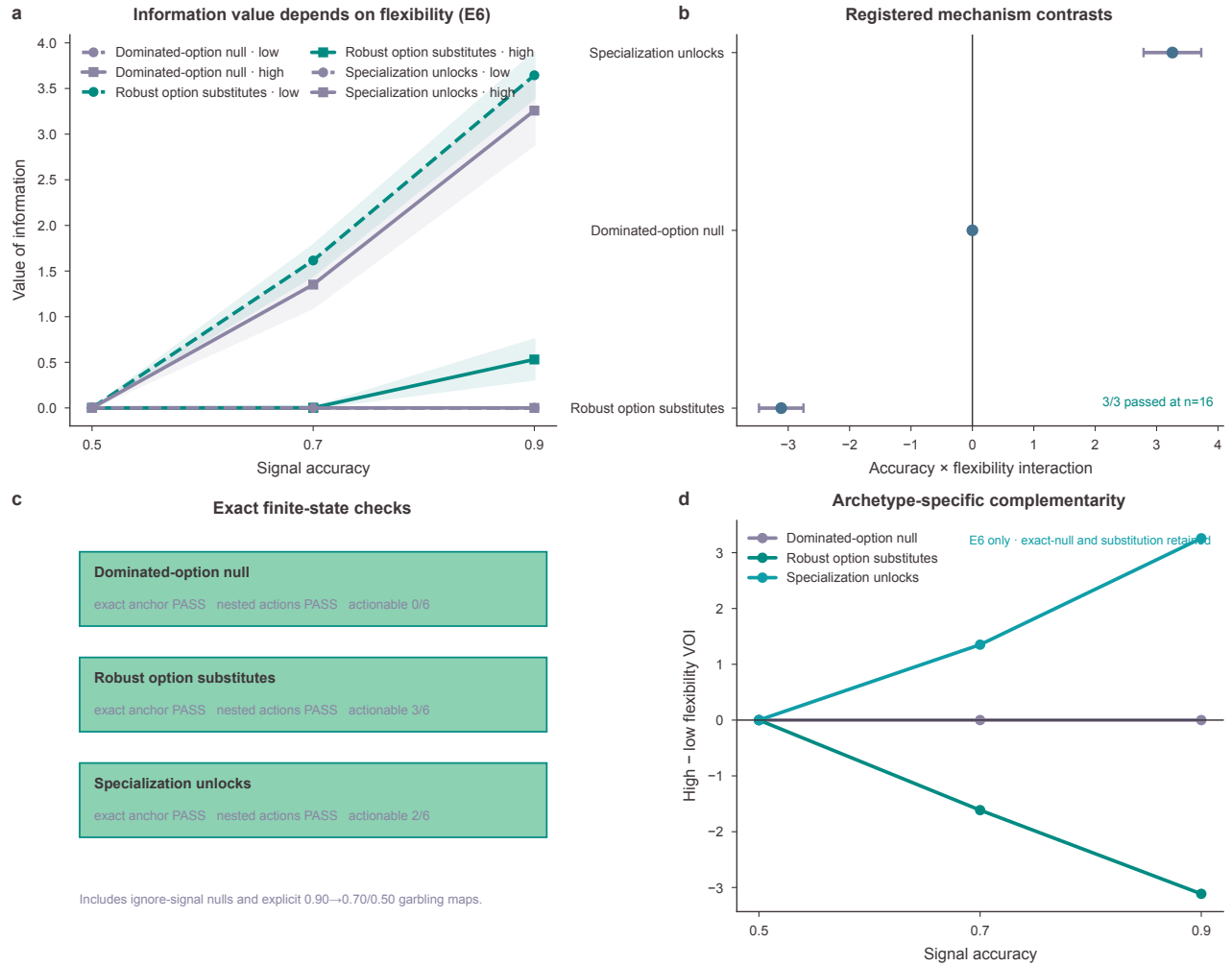


Figure 5: Information and action-set flexibility can complement, substitute or be unrelated. The E6-only display reports three registered archetypes, their interaction intervals, exact finite-state checks and archetype-specific flexibility gains. All three family-wise intervals meet the pre-specified precision criterion.

Aggregation changes the empirical statement. Operating-margin inversion intensity is 0.148 nationally and total-cost-margin inversion is 0.481. Relative yield is tied by construction in all 9 national years and is therefore non-informative rather than rank-aligned. The national summaries neither validate a preferred score nor negate state heterogeneity; they show that the spatial support of any ranking-reversal claim must be declared.

The evidence boundary is asymmetric. Public data identify acreage, yield and transparent accounting relations, and they support descriptive rank agreement and transition estimands. They do not reveal private objectives, constraints, beliefs, downside-risk limits or a selected optimum. Model-linked signatures are therefore reported as compatibility checks, not mechanism tests. The full annual, leave-one-state-out and sample-flow checks appear in Supplementary Fig. S5.

6 Discussion

The central result is that ranking reversal is a relationship, not a single mechanism. It compares an ordinal statement with a cardinal allocation drawn from a constrained optimal face. Exact theory

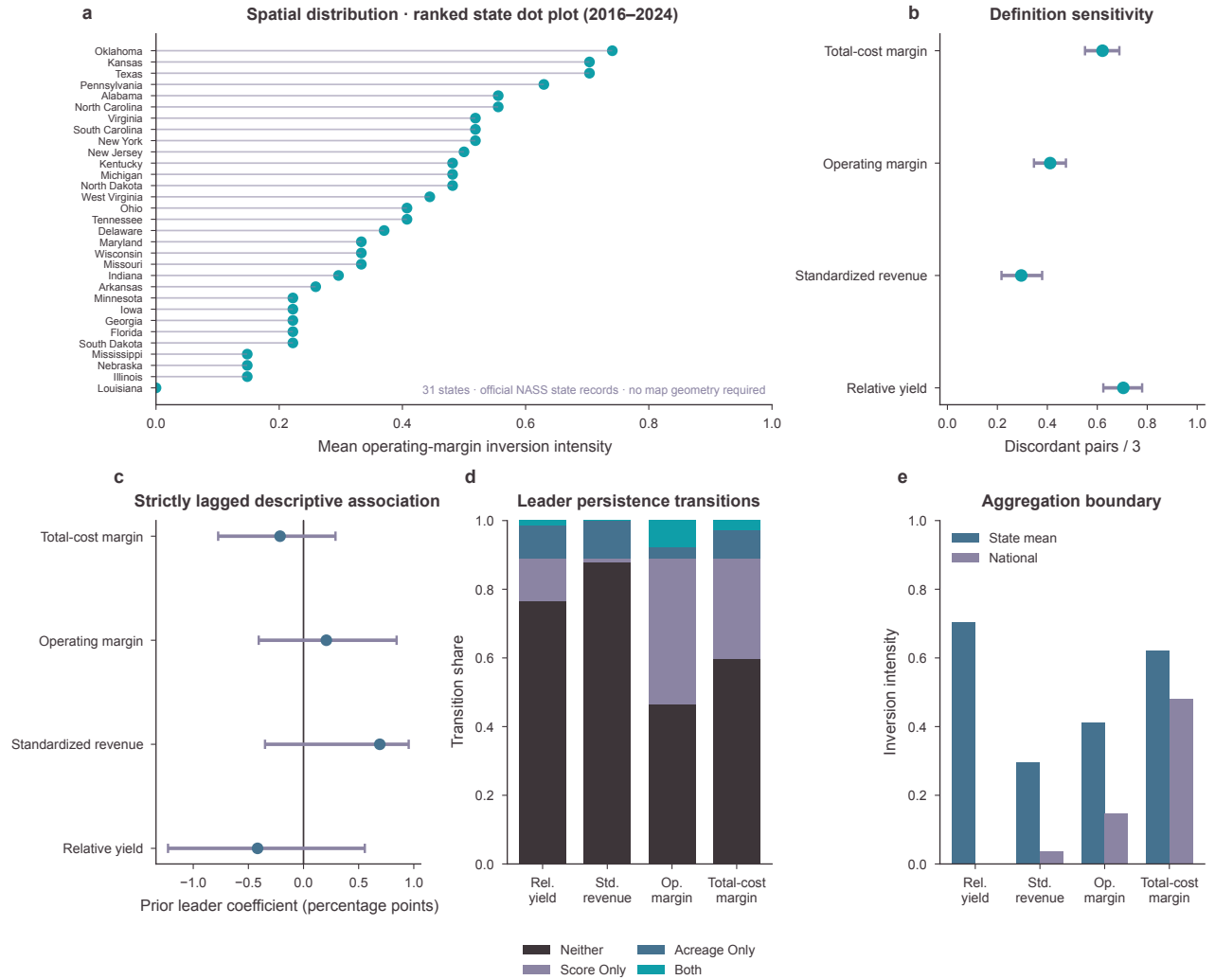


Figure 6: Spatial, definitional, temporal and aggregation boundaries in official data. The panel covers 248 complete state-years in 31 states from 2016 to 2024. The ranked state dot plot uses official state records; intervals use 5000 state-cluster draws. Every primary strictly lagged interval includes zero. Observed acreage is not interpreted as a model optimum.

establishes why the ranking alone is insufficient, and controlled experiments identify two positive mechanism classes: operational interventions can displace the ranked leader, and the interaction between information and action-set flexibility changes sign with the actions that flexibility enables.

This distinction avoids two interpretive extremes. Treating every score–acreage mismatch as an error ignores objectives, constraints and uncertainty. Treating every mismatch as uninformative overlooks controlled settings in which the intervention and optimal face are known. E2 supports an operational mechanism because the constraints are assigned and the entire optimal face is audited. E6 supports actionability because signal and action-set contrasts are factorial and exact checks pass. Neither experiment establishes population prevalence.

The inconclusive experiments are part of the result. E1, E3, E4 and E5 show that deterministic reproduction, solver agreement and visually structured contrasts do not guarantee the required precision. Preserving their intervals, stopping outcomes and infeasible rows prevents a passing sub-contrast from being promoted retrospectively. Numerical integrity and inferential precision remain separate criteria.

The extended public-data panel establishes a different evidence layer. Concurrent rank–acreage inversion is widespread but definition- and state-dependent; all primary lagged intervals include zero; and national aggregation changes the pattern. These data do not measure a farm’s feasible set or objective and cannot identify observed acreage as optimal. In particular, they support no inference about private downside-risk preferences, dependence mechanisms, welfare or regret relative to a latent optimum.

Several limitations remain. The empirical analysis covers three crops and uses national price and cost series to create transparent accounting scores, omitting local basis, heterogeneous costs and private expectations. The model is linear and single-period, excluding fixed charges, endogenous prices, spatial interaction and dynamic rotations. The simulation parameters are controlled design values rather than farm estimates. Future work requires decision-level observations on expectations, constraints and contracts before the controlled mechanisms can be connected to field prevalence.

The practical implication is procedural. A crop ranking intended to guide acreage should disclose the cardinal payoff mapping, operational constraints, joint uncertainty model, downside-risk convention and selection rule. When those objects are unavailable, score–acreage comparisons should remain descriptive. A prediction becomes prescriptive only through the decisions it can alter and the constraints under which those decisions are feasible.

7 Methods

7.1 Optimization model and optimal-face classification

Per-acre stochastic margins are combined linearly in acreage. Operational feasibility includes lower and upper crop bounds, land, budget and shared linear restrictions. Loss-CVaR is

$$r_\alpha(x) = \min_{v \in \mathbb{R}} \left\{ v + \frac{1}{1 - \alpha} \mathbb{E}[(-\tilde{\pi}^\top x - v)_+] \right\}, \quad (6)$$

with a free VaR auxiliary variable. In finite scenarios, non-negative excess-loss variables give a linear programme. Possible and universal reversal are computed by minimizing and maximizing pairwise acreage gaps over the objective-equivalent optimal face. Selected reversal uses the frozen deterministic selection rule. Direct loss-CVaR, primal feasibility, complementary slackness and stationarity are recomputed independently.

7.2 Confirmatory simulations

The E1–E6 design was frozen before execution. Each experiment specifies factors, estimands, multiplicity, sequential replication checks, precision targets and a ceiling. E2 crosses assigned operational interventions against a fixed ranking and law. E6 crosses signal information and action-set flexibility in three constructive environments. Promotion requires every registered primary interval to meet its experiment-level target. Registered infeasible rows are retained and classified. Reverse-order replay and solver-method sensitivity rerun named tasks.

7.3 Mechanism attribution

The M0–M4 path adds cardinal margins, operations, downside risk and dependence to one closed benchmark. Because sequential contributions depend on order, every block coalition is solved and Shapley contributions are calculated for allocation, expected margin, loss-CVaR and concentration. The efficiency residual compares the contribution sum with the full-minus-base difference. KKT pressure terms are evaluated from Eq. 5 and interpreted as local accounting terms.

7.4 Official data and frozen empirical design

USDA NASS Crop Production annual summaries provide state and national planted acreage and yield for corn, soybeans and winter wheat. Non-overlapping reports cover 2016–2024. The parser’s overlapping 2024 block reproduces the previously frozen table exactly. USDA ERS national harvest prices, operating costs and total costs are joined by crop-year; the registered BLS CPI-U series supplies the real-dollar conversion. A state-year is retained only when all three crops have acreage and yield. Missing and suppressed values are not imputed.

Four score definitions were frozen before extended results were inspected: relative yield, standardized revenue, operating margin and total-cost margin. Concurrent estimands are Kendall’s tau-b, Spearman correlation when defined, pairwise inversion intensity and top-rank disagreement. The temporal model regresses acreage-share change on prior score-leader status and prior acreage share with crop, year and state fixed effects. Scores strictly precede the outcome. Uncertainty uses 5000 state-cluster bootstrap draws. All empirical estimands are descriptive.

The official NASS and ERS series were audited forward, but the next year was excluded because a complete annual CPI-U value matching the frozen construction was unavailable at the design freeze. County reconstruction was not run because no targeted credential-free Quick Stats query was available and the public all-crops bulk snapshot was not a bounded extraction. Both decisions are recorded with source URLs, checksums and blockers; no county claim is made.

7.5 Visualization and reproducibility

The central colour card is #3D3539, #0F9EA8, #008B82, #45728F, #8CD1B2 and #8B84A3. Corn, soybean and winter wheat are fixed to #45728F, #008B82 and #8CD1B2; positive evidence uses #0F9EA8, unresolved evidence uses #8B84A3, and text and axes use #3D3539. An automated validator rejects off-card SVG colours and embedded raster panels. Each figure is exported as editable SVG/PDF, 300-dpi PNG and 600-dpi TIFF and inspected at full size, 183 mm, 89 mm, grayscale, deuteranopia and protanopia.

Raw sources, tidy outputs, source tables, figures and manuscript numbers carry SHA-256 lineage. The empirical pipeline is accepted only when two isolated runs are byte-identical. The manuscript and Supplementary Information are compiled in isolated directories; citations, references, fonts, page renders and archive checksums are checked before release.

Declarations

Acknowledgements. To be completed by the authors before submission.

Author contributions. To be completed using the journal’s CRediT format.

Competing interests. The authors must confirm the final statement before submission.

Ethics. The study uses public aggregate data and no human-participant or animal data.

Data availability. All admitted source identifiers, immutable raw-file checksums, processing contracts and derived analysis files are recorded in the repository’s data and evidence registries. Public-source redistribution remains subject to the originating agencies’ terms.

Code availability. The complete theory, optimization, simulation, empirical, visualization and manuscript pipelines are versioned in the companion repository. Reproduction commands and checksums are provided with each artifact class.

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